3. Preliminaries

Normal-Form Game. A two-player normal-form game is defined as a tuple $(N, \mathcal{A}, \mathbf{w})$, where $N = \{1, 2\}$ is a set of two players, indexed by i, $\mathcal{A} = \mathcal{A}_1 \times \mathcal{A}_2$ is the joint action space, and $\mathbf{w} = (w_1, w_2)$ with $w_i : \mathcal{A} \to \mathbb{R}$ is a reward function for the player i. In a two-player common payoff game, two-player rewards are the same, meaning $w_1(a_1, a_2) = w_2(a_1, a_2)$ for $a_1, a_2 \in \mathcal{A}$.

Open-Ended Meta-Games and Analysis. A meta-game is characterized by the tuple $(\mathcal{N}, \mathcal{S}, \mathcal{M})$, where $\mathcal{N}$ represents the set of players, $\mathcal{S}$ is a set of policies (like a set of RL models), and $\mathcal{M}$ designates the meta-game payoff matrix. In common payoff cooperative meta-game with two players, the meta-game payoff is $\mathcal{M} = \left\{\phi(S_1, S_2) : (S_1, S_2) \in \mathcal{S}^1 \times \mathcal{S}^2\right\}$, where $\phi : \mathcal{S}^1 \times \mathcal{S}^2 \to \mathbb{R}$ is the utility function. The meta-game payoff is a symmetric matrix. The primary distinction between a meta-game and a normal-form game lies in their respective actions. In the case of a meta-game, a strategy such as a DRL model constitutes an action, as opposed to the atomic actions (e.g., up and down) typical of normal-form games. Open-ended meta-game refers to the continual learning process, wherein new strategies are persistently generated and incorporated into the strategy sets throughout the training phase.

Empirical Game-Theoretic Analysis (EGTA) is the study of finding meta-strategies based on experience with prior strategies (Walsh, Das, Tesauro, & Kephart, 2002; Tuyls, Pérolat, Lanctot, Leibo, & Graepel, 2018). An empirical game is built by discovering strategies and meta-reasoning about exploring the strategy space (Lanctot et al., 2017). Furthermore, empirical gamescapes (EGS) are defined as the convex hull of the payoff vectors of all strategies (Balduzzi et al., 2019). Given a population $\mathcal{N}$ of n strategies, the empirical gamescapes is often defined as

$$\begin{aligned} \mathcal{G} &= \{\text{convex mixture of rows of } \mathcal{M}\} \\ &= \left\{ \sum_i \alpha_i \cdot \boldsymbol{m}_i : \alpha \geq \mathbf{1}, \alpha^{\mathbf{T}} \mathbf{1} = \mathbf{1}, \mathbf{m_i} = \mathcal{M}_{[\mathbf{i},:]} \right\} \end{aligned} \tag{1}$$

where $\mathcal{M}$ is the empirical payoff matrix with the expected outcomes for each joint strategy.

Cooperative Theoretic Concepts. In the present study, our main focus is on characteristic function games, a particular class of cooperative games in which a coalition's generated value is represented by a characteristic function (Chalkiadakis, Elkind, & Wooldridge, 2011). We consider a set of players $\mathcal{N} = \{1, \ldots, n\}$, where a coalition is denoted as a subset of players $\mathcal{N}$, symbolized by $C \subseteq \mathcal{N}$. The player set $\mathcal{N}$ is also known as the grand coalition. A characteristic function game, denoted by G, consists of a pair (N, v), where $N = \{1, \ldots, n\}$ represents a finite, non-empty set of agents, and $v : 2^N \to \mathbb{R}$ is the characteristic function. This function assigns a real number $v(C)$ to each coalition $C \subseteq N$, with the number $v(C)$ typically considered as the coalition's value.

In addition, in this study, we examine transferable utility games (TU games), which are based on the underlying assumption that the coalitional value $v(C)$ can be divided among the members of C in any way agreed upon by the members of the coalition. This assumption allows for greater flexibility in analyzing cooperative problem solving and the distribution of the resulting payoffs.

Shapley Value (Shapley, 1971) is one of the important solution concepts for characteristic function games (Chalkiadakis et al., 2011; Peleg & Sudhölter, 2007). The Shapley Value aims to distribute fairly the collective value, like the rewards and cost of the team across individuals by each player's contribution. Taking into account a coalition game $(\mathcal{N}, v)$ with a strategy set $\mathcal{N}$ and characteristic function v, the Shapley Value of a player $i \in \mathcal{N}$ could be obtained by

$$SV(i) = \frac{1}{n!} \sum_{\pi \in \Pi_{\mathcal{N}}} v(P_i^\pi \cup \{i\}) - v(P_i^\pi), \tag{2}$$

where π is one of the one-to-one permutation mappings from $\mathcal{N}$ to itself in the permutation set Π and $\pi(i)$ is the position of player $i \in \mathcal{N}$ in permutation π. $P_i^\pi = \{j \in \mathcal{N} | \pi(j) < \pi(i)\}$ is the set of all predecessors of i in π.

Graph Theoretic Concepts. In graph theory, a weighted directed graph or network can be represented as $D = (V, E, w)$, consisting of a set of vertices V, a set of directed edges E, and a weight function $w : E \to \mathbb{R}^+$ that assigns positive real numbers to each edge. If the graph has the property that $(v, u) \in E$ entails $(u, v) \in E$ and $w(v, u) = w(u, v)$ for all $(u, v) \in E$, it can be considered an undirected weighted graph, also known as a weight graph. An unweighted graph is a special case of a weighted graph, where the weight function assigns a value of 1 to every edge $(u, v) \in E$.

Node Centrality is a graph theory concept that quantifies a node's relative importance or influence within a network. It is a measure of how central a node is to the overall network structure, with higher centrality values indicating greater importance. Degree Centrality is one of the simplest centrality concepts, based on the number of edges connected to a node (Freeman, 1978). In directed graphs, it can be further divided into in-degree (number of incoming edges) and out-degree (number of outgoing edges) Centrality.

PageRank (Page, Brin, Motwani, & Winograd, 1999) is a centrality measure used to rank web pages in search engine results by estimating the relative importance of each page in a hyperlink network. PageRank(WPG) (Xing & Ghorbani, 2004) is an extension of the original PageRank algorithm that considers edge weights in addition to the basic structure of the network. This modification allows the algorithm to handle networks where the strength of connections between nodes varies, such as in citation networks or social networks where the influence of nodes may differ. The formula of WPG is given as follows:

$$\sigma(u) = (1 - d) + d \sum_{v \in B(u)} \sigma(v) \frac{I_u}{\sum_{p \in R(v)} I_p} \frac{O_u}{\sum_{p \in R(v)} O_p}, \tag{3}$$

where d is the damping factor set to 0.85, $B(u)$ is the set of nodes that point to u, $R(v)$ denotes the nodes to which v is linked, and I, O are the degrees of inward and outward of the node, respectively.

4. Cooperative Open-Ended Learning

4.1 Graphic-Form Games (GFGs)

It is important to evaluate cooperative incompatibility and identify those failed-to-collaborate strategies to conquer cooperative incompatibility. Therefore, we propose graphic-form games

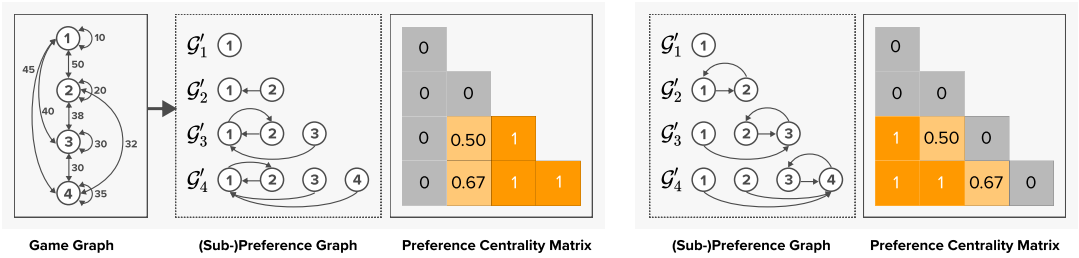

Figure 1: The Game Graph, (sub-) preference graph and corresponding preference centrality matrix. The (sub-) preference graphs are for all four iterations in the training process, and the corresponding preference in-degree centrality matrix is based on them. As can be observed in $\mathcal{G}'_3$ and $\mathcal{G}'_4$ in *(a)*, the newly updated strategies fail to be preferred by others and have centrality values of 1, despite an increase in the mean of rewards with all others. In *(b)*, we illustrate an ideal learning process in which a newly generated strategy can achieve higher outcomes than all previous strategies.

(GFGs) to reformulate normal-form cooperative games from the perspective of game theory and graph theory, which is the natural development of empirical games (Balduzzi et al., 2019). The definition of GFG is given below.

Definition 4.1 (Graphic-Form Game)**.** Let a collection of strategies $\mathcal{N} = \{1, 2, \cdots, n\}$ be given, where each strategy could be a parameterized network or a human player. A two-player graphic-form game (GFG) can be defined as a triplet $\mathcal{G} = (\mathcal{N}, \mathbf{E}, \mathbf{w})$, which can be represented as a weighted directed graph. Here, $\mathcal{N}$, $\mathbf{E}$, and $\mathbf{w}$ represent the sets of nodes, edges, and weights, respectively. For a given edge (i, j), $\mathbf{w}(i, j)$ denotes the anticipated outcome when strategy i competes against strategy j. The visual representation of a GFG is referred to as a game graph.

The payoff matrix of $\mathcal{G}$ is denoted as $\mathcal{M}$, where $\mathcal{M}(i, j) = \mathbf{w}(i, j), \forall i, j \in \mathcal{N}$. Our goal is to improve the upper bound of other strategies' outcomes in the cooperation within the population, which implies that the strategy should be preferred over other strategies.

Moreover, we propose preference graphic-form games (P-GFGs) as an efficient tool to analyze the current learning state, which can profile the degree of preference for each node in GFGs. Specifically, P-GFG is a subgraph of GFG, where each node only retains the out-edge with maximum weight among all out-edges except for its self-loop. Given a GFG $(\mathcal{N}, \mathbf{E}, \mathbf{w})$, the P-GFG could be defined as $\mathcal{G}' = \{\mathcal{N}, \mathbf{E}', \mathbf{w}\}$, where $\mathbf{E}' = \{(i, j) | \arg\max_j \mathbf{w}(i, j), \forall j \in \{\mathcal{N} \backslash i\}, \forall i \in \mathcal{N}\}$ is the set of edges. The graphic representation of P-GFG is called a preference graph. The game graph delineates the interaction of players within a weighted directed graph, in which the weight signifies the payoff or utility derived from two participating agents. However, the preference graph pertains to the depiction of each agent identifying the partner with whom they can achieve the most significant utility in the population. This